

PATNA, India

WMO: 424920

Lat: 25.591N

Lon: 85.088E

Elev: 52

StdP: 100.70

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.7	8.9	3.9	5.0	26.2	5.4	5.6	20.4	5.7	20.3	5.0	20.0	0.4	270	0.354

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	8.7	41.3	23.2	39.9	23.4	38.1	24.2	29.1	34.1	28.7	33.5	28.4	32.8	3.3	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.0	24.3	31.2	27.6	23.7	30.8	27.2	23.1	30.5	95.8	34.3	94.0	33.7	92.4	32.9	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.7	5.1	DB	5.3	43.7	1.5	1.4	4.3	44.7	3.4	45.5	2.5	46.3	1.4	47.3	(4)
			WB	5.1	30.1	1.4	0.6	4.1	30.5	3.3	30.8	2.6	31.1	1.6	31.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.0	15.6	19.9	25.3	30.3	31.8	32.0	30.1	30.0	29.4	27.3	22.6	17.4	(6)
(7)		DBStd	5.94	2.61	2.47	2.74	2.18	2.40	2.46	1.61	1.33	1.55	1.94	2.13	2.41	(7)
(8)		HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	144	89	11	0	0	0	0	0	0	0	0	0	44	(9)
(10)		CDD10.0	5841	174	277	475	608	676	659	623	620	583	537	379	230	(10)
(11)		CDD18.3	2943	4	54	217	358	418	409	364	362	333	278	129	16	(11)
(12)		CDH23.3	37492	38	457	2354	4946	6201	6026	4798	4718	4124	2800	906	126	(12)
(13)	CDH26.7	19647	1	102	1178	2971	3857	3663	2361	2266	1847	1139	257	6	(13)	
(14)	Wind	WSAvg	1.8	1.1	1.4	1.8	2.4	2.8	2.7	2.4	2.2	2.0	1.0	0.6	0.7	(14)
(15)	Precipitation	PrecAvg	1169	13	12	13	10	42	122	367	271	219	89	8	5	(15)
(16)		PrecMax	1886	31	69	55	40	132	302	801	475	503	373	70	32	(16)
(17)		PrecMin	580	0	0	0	0	0	2	64	71	26	0	0	0	(17)
(18)		PrecStd	324	11	16	17	12	34	82	179	116	128	100	17	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	31.2	38.0	42.2	43.1	43.1	37.0	35.8	35.2	34.9	31.8	27.2	(19)
(20)		0.4%	MCWB	17.6	20.3	20.7	21.4	22.0	23.6	28.1	28.2	28.0	26.6	23.1	19.1	(20)
(21)		2%	DB	24.2	29.2	35.8	40.2	41.8	41.0	35.5	34.8	34.4	33.6	30.3	26.0	(21)
(22)		2%	MCWB	16.6	19.4	20.8	21.3	22.6	25.6	27.8	28.1	27.8	26.1	21.9	18.0	(22)
(23)		5%	DB	23.1	27.9	34.1	39.0	40.1	39.0	34.5	34.0	33.8	32.8	29.5	24.9	(23)
(24)		5%	MCWB	15.9	18.8	20.8	21.6	23.5	26.5	27.8	28.0	27.6	25.6	21.6	17.3	(24)
(25)		10%	DB	21.8	26.2	32.8	37.2	38.1	37.1	33.5	33.1	32.9	31.8	28.2	23.2	(25)
(26)		10%	MCWB	15.5	17.9	20.5	21.9	24.8	27.0	27.8	27.8	27.4	25.0	20.8	16.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.3	22.2	24.7	27.3	28.8	29.5	29.4	29.5	29.3	28.5	25.4	20.9	(27)
(28)		0.4%	MCDWB	23.4	28.2	32.0	34.7	36.4	35.7	33.8	33.4	33.5	32.4	29.0	25.0	(28)
(29)		2%	WB	18.1	20.9	23.4	26.4	28.0	28.9	28.8	28.9	28.6	27.5	23.7	19.6	(29)
(30)		2%	MCDWB	22.1	27.2	30.8	33.8	35.7	35.6	33.1	32.9	32.7	31.2	28.0	23.8	(30)
(31)		5%	WB	17.2	19.8	22.4	25.6	27.4	28.5	28.4	28.5	28.2	26.8	22.7	18.6	(31)
(32)		5%	MCDWB	21.1	25.7	30.1	32.8	34.6	34.6	32.5	32.2	31.8	30.5	27.1	22.7	(32)
(33)		10%	WB	16.2	19.0	21.4	24.8	26.8	28.0	28.1	28.1	27.8	26.2	22.0	17.7	(33)
(34)		10%	MCDWB	20.0	24.5	29.7	31.9	33.6	33.5	31.9	31.5	31.1	29.7	26.5	21.6	(34)
(35)	Mean Daily Temperature Range		MDBR	11.1	12.6	13.9	13.6	11.4	8.7	5.9	5.8	6.2	8.9	12.1	12.0	(35)
(36)		5% DB	MCDDBR	13.5	13.9	15.1	16.1	14.4	12.0	7.5	6.9	7.4	9.3	12.0	13.6	(36)
(37)			MCWBR	6.5	5.6	4.9	5.3	4.5	3.6	2.5	2.3	2.7	3.3	4.8	6.3	(37)
(38)			MCDDBR	11.3	12.5	13.3	12.0	10.9	9.2	6.6	6.3	6.7	7.9	10.7	11.5	(38)
(39)		5% WB	MCWBR	6.1	5.8	5.2	5.0	4.3	3.5	2.7	2.6	2.8	3.3	4.9	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.597	0.548	0.545	0.622	0.708	0.723	0.622	0.589	0.589	0.622	0.690	0.676	(40)	
(41)		taud	1.658	1.810	1.850	1.740	1.616	1.620	1.839	1.940	1.914	1.768	1.572	1.560	(41)	
(42)		Ebn at Noon	639	730	764	717	656	642	710	732	719	659	561	551	(42)	
(43)		Edh at Noon	221	203	205	234	264	262	210	189	190	209	240	237	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.12	4.46	5.74	6.47	6.38	5.46	4.49	4.70	4.60	4.52	3.80	3.02	(44)	
(45)		RadStd	0.37	0.30	0.19	0.30	0.32	0.37	0.39	0.23	0.33	0.32	0.19	0.35	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page